

1. a) let,

the total time required by Shafi = y ns

So, " " " " " Rafi = $(y - 50)$ ns

Now,

For Shafi,

$$100 + \frac{1024 \times 10^3}{x \times 1024} + 200 = y \dots (1)$$

For Rafi,

$$100 + \frac{1024 \times 10^3}{2x \times 1024} + 200 = y - 50 \dots (2)$$

$$(1) - (2) \Rightarrow \frac{1024 \times 10^3}{1024x} - \frac{1024 \times 10^3}{1024 \times 2x} = y - y + 50$$

$$\Rightarrow \frac{1000}{x} - \frac{1000}{2x} = 50$$

$$\Rightarrow \frac{1000}{x} - \frac{500}{x} = 50$$

$$\Rightarrow \frac{(1000 - 500)}{x} = 50 \Rightarrow x = \frac{500}{50} = 10$$

$$\therefore \boxed{x = 10}$$

b) Command Register = 95 H = 1001 0101

So, Port A is configured as output port. $\downarrow$ Port A (Output)

That is why, we cannot take input from any pins of Port A.
So, the system will not work correctly.

$$\begin{array}{r}
 1 \dots 8A32 \\
 + \quad DFED \\
 \hline
 16A1F \\
 \text{(i)} \quad \begin{array}{r} 0110 \quad \underline{0001} \quad 1111 \end{array}
 \end{array}$$

(2) OF = 1 [Adding 2 negative $\rightarrow$ positive]

PF = 0 [odd 1's]

$$\begin{array}{r}
 \text{(ii)} \quad 6A1F \\
 \quad \quad \underline{E321} \\
 \quad \quad A6FE
 \end{array}$$

CF = 1 [small no - large no]

AF = 0 [No borrow from bit 4 to 3]

2(b) : False.

Segment's first address rightmost

~~bit~~ hex digit is always 0.

$$S.A = \text{segment} \times 10^4$$

3) a) I) Physical address (PA) = 2BCEC

For DS:

$$PA = DS * 10 + \text{offset}_1$$

$$2BCEC = 1E5B * 10 + \text{offset}_1$$

$$\Rightarrow 2BCEC - 1E5B0 = \text{offset}_1$$

$$\Rightarrow \text{offset}_1 = D73C < FFFF \quad \text{valid}$$

For SS:

$$PA = SS * 10 + \text{offset}_2$$

$$\Rightarrow 2BCEC = 2F1D * 10 + \text{offset}_2$$

$$\therefore \text{offset}_2 = FFFF CB1C \quad [(-ve)]$$

Stack Segment's starting location > PA

So, invalid offset $[\because 2F1D0 > 2BCEC]$

For ES:

$$PA = ES * 10 + \text{offset}_3$$

$$\Rightarrow 2BCEC = 2BCE * 10 + \text{offset}_3$$

$$\Rightarrow \text{offset}_3 = 000C < FFFF \quad \text{valid}$$

II) DS:

starting location = 1E5B0

ending location = 1E5B0 + FFFF
= 2E5AF

SS:

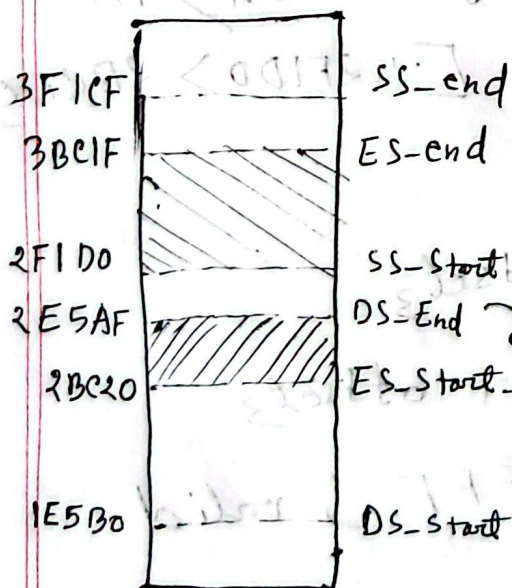
starting location = 2F1D0

ending location = 2F1D0 + FFFF
= 3F1CF

ES:

starting location = 2BC20

ending location = 2BC20 + FFFF
= 3BC1F



Overlapping locations between
SS and ES = 3BC1F - 2F1D0 + 1
= CA50H = 51792D

Overlapping locations between
ES and DS = 2E5AF - 2BC20 + 1
= 2990H = 10640D

3) b)

Let, capacities of SRAM, DRAM₁, DRAM₂
are x , $2x$, $4x$ MB

$$\therefore x + 2x + 4x = 28$$

$$\Rightarrow x = 4$$

$\therefore$ For SRAM:

$$\frac{2^{n_1} \times 16}{8 \times 2^{20}} = 4$$

$$\Rightarrow 2^{n_1} = 2^{21}$$

$$\therefore n_1 = 21$$

For DRAM₁:

$$\frac{2^{n_2} \times 32}{8 \times 2^{20}} = 8$$

$$\therefore 2^{n_2} = 2^{21}$$

$$\therefore n_2 = 21$$

For DRAM₂:

$$\frac{2^{n_3} \times 64}{8 \times 2^{20}} = 16$$

$$\Rightarrow 2^{n_3} = 2^{21}$$

$$\Rightarrow n_3 = 21$$

All of their address buses are 21 bits.

$$\text{RAM Capacity} = 2'' \times 16 \text{ bits} = 4 \text{ KB}$$

$$\# \text{ RAMs} = \left\lceil \frac{18 \text{ KB}}{4 \text{ KB}} \right\rceil = 5$$

Second RAM starts at = 2D800H

First RAM ending address = 2D7FFH

Since 11bit Address Bus,

$$\text{First RAM starting addr} = 2D7FFH - 7FFH = 2D000H$$

Addr. Mapping

RAM-0 : 2D000H - 2D7FFH

RAM-1 : 2D800H - 2DFFFH

RAM-2 : 2E000H - 2E7FFH

First Address
(2D000H)

A_{17}	A_{16}	A_{15}	A_{14}	A_{13}	A_{12}	A_{11}	$A_{10}-A_0$
1	0	1	1	0	1	0	0...0
1	0	1	1	0	1	0	1...1

